

Lesson plan (Tentative)

Name of the Programme:	B.Tech (Hons.)	
Title of the Course:	Compiler Design	
Course code	CS3704	
Semester:	VI	
Names of the Course Instructor(s)	Dr. Ananya Chakraborty, Dr Sangeeta Bhanja Chaudhuri, Prof Shoeb Ahmad, Prof T Narmadha, Prof Kalpana Devi S	
Course credit	No. of hours per week (3+0+0)	Total no. of Teaching hours
03	03 Hours	45 Hours

Sessio n	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	I	Introduction to Compiler	Chalk & Talk, real-world examples of translators	Aho, Lam, Sethi & Ullman, <i>Compilers: Principles, Techniques and Tools</i> , Ch.1
2.	I	Compiler and Interpreter	PPT presentation, comparison discussion	Aho et al., Ch.1
3.	I	Language Processing System	Diagram explanation, flowchart activity	Aho et al., Ch.1
4.	I	Phases of a Compiler	PPT with block diagram, short quiz	Aho et al., Ch.1
5.	I	Role of Lexical Analyzer	Token identification exercise	Aho et al., Ch.3
6.	I	Regular Expressions	Problem solving, pattern matching tasks	Aho et al., Ch.3
7.	I	Finite Automata (DFA & NFA)	Worked examples, board problems	Hopcroft & Ullman, <i>Introduction to Automata Theory</i> , Ch.3
8.	I	Conversion of FA to Regular Expression	Worked examples	Aho et al., Ch.3
9.	I	Bootstrapping and Lexical Analyzer Generator (LEX)	Tool demonstration +PPT	Aho et al., Ch.3
10.	II	Context Free Grammar (CFG)	Chalk & talk, grammar examples	Aho et al., <i>Compilers</i> , Ch.4
11.	II	Derivations and Parse Trees	Worked derivations, board illustration	Aho et al., Ch.4
12.	II	Ambiguity and Eliminating Ambiguity	Expression grammar case study	Aho et al., Ch.4

13.	II	Left Recursion and Left Factoring	Grammar transformation exercises	Aho et al., Ch.4
14.	II	Top-Down Parsing: Backtracking & Recursive Descent	Algorithm walkthrough	Aho et al., Ch.4
15.	II	Predictive Parsing & LL(1) Grammars	FIRST/FOLLOW computation, table construction	Aho et al., Ch.4
16.	II	Bottom-Up Parsing & LR Parsers	Shift-reduce parsing demo	Aho et al., Ch.5
17.	II	SLR, CLR and LALR Parsers	Parsing table construction & comparison	Aho et al., Ch.5
18.	II	Operator Precedence Parsing	Precedence table creation & parsing example	Aho et al., Ch.4
19.	III	Syntax-Directed Definition (SDD)	Chalk & talk, attribute explanation	Aho et al., <i>Compilers</i> , Ch.6
20.	III	Construction of Syntax Trees	Board illustration, examples	Aho et al., Ch.6
21.	III	S-Attributed Definitions	Worked attribute evaluation examples	Aho et al., Ch.6
22.	III	L-Attributed Definitions	Step-by-step dependency graph	Aho et al., Ch.6
23.	III	Syntax-Directed Translation Schemes	Grammar-based translation activity	Aho et al., Ch.6
24.	III	Applications of Syntax-Directed Translation	Case study discussion	Aho et al., Ch.6
25.	III	Intermediate Code & Postfix Notation	Expression translation problems	Aho et al., Ch.6
26.	III	Parse Trees vs Syntax Trees	Comparative analysis	Aho et al., Ch.6
27.	III	Translation of Simple Statements control flow statements.	Example-based teaching	Aho et al., Ch.6
28.	IV	Introduction to Symbol Tables	Chalk & talk, real-time examples	Aho et al., <i>Compilers</i> , Ch.7
29.	IV	Data Structures for Symbol Tables	Comparative discussion (array, list, hash table, tree)	Aho et al., Ch.7
30.	IV	Operations on Symbol Tables	Insert, lookup, delete – worked examples	Aho et al., Ch.7

31.	IV	Representing Scope Information	Block structure illustration, scope tree	Aho et al., Ch.7
32.	IV	Run-Time Environments – Overview	Concept explanation with diagrams	Aho et al., Ch.7
33.	IV	Simple Stack Allocation Scheme	Activation record layout, examples	Aho et al., Ch.7
34.	IV	Lexical Phase Errors	Token error examples, discussion	Aho et al., Ch.3
35.	IV	Syntactic and Semantic Errors	Error cases and recovery techniques	Aho et al., Ch.4 & Ch.6
36.	IV	Error Detection & Recovery Strategies	Panic-mode and phrase-level recovery	Aho et al., Ch.4
37.	V	Design Issues in Code Generation	Chalk & talk, discussion	Aho et al., <i>Compilers</i> , Ch.8
38.	V	Target Language & Addressing Modes	Instruction format explanation	Aho et al., Ch.8
39.	V	Addresses in the Target Code	Worked examples	Aho et al., Ch.8
40.	V	Basic Blocks and Flow Graphs	CFG construction exercise	Aho et al., Ch.9
41.	V	Optimization of Basic Blocks	Common sub-expression elimination	Aho et al., Ch.9
42.	V	Machine-Independent Optimizations	Code improvement problems	Aho et al., Ch.9
43.	V	Loop Optimization Techniques	Loop invariant motion examples	Aho et al., Ch.9
44.	V	DAG Representation, Value Numbering & Algebraic Laws	Diagram-based problem solving	Aho et al., Ch.9
45.	V	Data-Flow Analysis: Constant Propagation & Liveness Analysis	Step-by-step analysis	Aho et al., Ch.9

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz (Pen and Paper)	20 marks out of 70
CIE-2 - Theory (Pen and Paper)	25 marks out of 70

CIE-3	10 marks out of 70 15 marks out of 70
- Case Study - Quiz (Pen and Paper)	

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2
CO 1	Understand and explore the fundamental phases of compiler design, including lexical analysis, syntax analysis, and semantic analysis.	3	2		2				2				
CO 2	Apply top-down and bottom-up parsing techniques, such as recursive descent, LL(1), and LR parsers, for syntactic analysis.	3	3	2	2				2				

CO 3	Develop syntax-directed translation schemes and generate intermediate code, while understanding runtime structures like symbol tables and their implementation.	3	3	3	3		1	2	2				
CO 4	Apply IP and Web security principles to design and evaluate secure systems in real-world networking scenarios.	3	3	2	3	1	1	2	2		3		

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Quiz	Quiz will be conducted for 20 marks.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Seminar	Video presentation = 10 marks (<i>Refer Table-1</i>)

	Quiz	Quiz (Pen and paper) = 15 marks (<i>Refer Table-2</i>)
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Table-1

Case study- Video Presentation (Max: 10 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1.	Topic Selection (1 Mark)	Evaluation of relevance and alignment of the chosen topic with the Compiler Design syllabus	Topic is highly relevant and clearly aligned with compiler design concepts. (1 Mark)	Topic is relevant but limited in scope or depth. (0.5 Mark)	Topic is not aligned with Compiler Design syllabus. (0 Mark)
2.	Understanding (3 Marks)	Assessment of correctness and completeness of explanation of compiler design concepts in the video	Clear, accurate, and complete explanation with appropriate examples. (3 Marks)	Mostly correct explanation with minor conceptual gaps. (2 Marks)	Incorrect, incomplete, or weak understanding. (1 Mark)
3	Content Quality (3 Marks)	Evaluation of depth of analysis, technical accuracy, and clarity of concepts presented	Strong analytical depth with clear and accurate concepts. (3 Marks)	Adequate analysis but lacks depth or clarity in parts. (2 Marks)	Superficial analysis with major conceptual errors. (1 Mark)
4	Presentation Skills (3 Marks)	Evaluation of speaking clarity, logical flow, timing, and overall organization of the video	Clear speech, well-structured flow, and effective time management. (3 Marks)	Understandable presentation with minor issues in flow or timing. (2 Marks)	Poor clarity, disorganized flow, or ineffective delivery. (1 Mark)

Table-2

Quiz-Pen and Paper (Total: 15 Marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Conceptual Understanding	Objective and short-answer questions covering core topics	Demonstrates thorough and accurate understanding of concepts; answers are	Demonstrates fair understanding with minor conceptual errors or omissions (4–5 Marks)	Demonstrates weak or incorrect understanding of concepts (1–3 Marks)

			precise and complete (6 Marks)		
2	Application of Concepts	Problem-based / analytical questions	Correctly applies concepts to solve all problems with appropriate logic and steps (5 Marks)	Applies concepts partially; minor logical or procedural errors (3–4 Marks)	Unable to apply concepts effectively; answers are largely incorrect or incomplete (1–2 Marks)
3	Accuracy, Clarity, and Presentation	Evaluation of written responses	Answers are well-structured, clearly presented, and logically sequenced (4 Marks)	Answers are mostly clear but lack structure or contain minor inaccuracies (3 Marks)	Poorly structured answers with significant inaccuracies or unclear presentation (1–2 Marks)

Signature of the Course Faculty

Signature of the Program Director